

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 10: Presenting Our Final Explanation and Sharing Seedlings with Kitale

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	A	B	D	A	B	A	A

USE AFTER: PREDICT PHASE — Think-Pair-Share: revisiting my Lesson 1 (Day 1) model

Q1. Mr. Wafula planted maize and beans on the same day, in the same field, after the same 25 mm of rain. Fewer bean seeds than maize seeds germinated, and they grew differently. Which explanation is best supported by the evidence?

- A. Healthy-looking seeds always germinate, so the differences are just chance.
- B. The beans were planted on a different day from the maize.
- C. The differences come from seed structure, seed condition (such as dormancy and viability) and the environment (such as waterlogging and oxygen).**
- D. The rain fell more on the beans than on the maize.

Correct answer: C. Both crops had the same planting day and rain, so those cannot explain the differences. Seed structure, seed condition and environment (for example a waterlogged patch with no oxygen) explain them.

USE AFTER: OBSERVE PHASE — Gallery Walk: checking that poster numbers match the Mr. Wafula germination data

Q2. Out of 100 Rosecoco bean seeds planted by Mr. Wafula, 80 germinated. The seeds that failed were: 8 with hard seed coats (dormant), 6 in the waterlogged patch, 4 non-viable and 2 damaged by pests. How many seeds failed to germinate in total, and does this agree with the 80 that germinated?

- A. 20 failed, which agrees with 80 germinating**
- B. 22 failed, so 78 should have germinated
- C. 12 failed, which agrees with 88 germinating
- D. 18 failed, so 82 should have germinated

Correct answer: A. $8 + 6 + 4 + 2 = 20$ failures, and $100 - 20 = 80$ germinated, so the causes account for all the failures. The figure 12 is the maize total, not the bean total.

USE AFTER: OBSERVE PHASE — Gallery Walk: the conditions tests (jar results 18, 0, 1, 0)

Q3. A learner puts 20 bean seeds in each of two jars kept at 25 °C. Jar 1 has moist cotton wool and air, and 18 seeds germinate. Jar 2 has the same temperature and water but the seeds are fully covered in water with no air, and only 1 seed germinates. What does this show?

- A. Seeds need soil to germinate.
- B. Seeds need oxygen for respiration to germinate, which is why waterlogged seeds fail.**
- C. Seeds germinate best when they have too much water.
- D. Seeds need light to germinate.

Correct answer: B. Only the oxygen supply differed between the two jars. Germination needs energy from respiration, which needs oxygen, so seeds in waterlogged soil fail.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: why 8 healthy-looking beans did not germinate

Q4. Eight bean seeds looked healthy but did not germinate in moist, warm soil. After the hard seed coat was scratched (scarification), 7 of the 8 germinated. What is the best explanation?

- A. The seeds were non-viable (dead) from the start.
- B. The seeds had too much oxygen before they were scratched.
- C. The seeds were damaged by pests, which scarification repaired.
- D. The seeds were dormant because the hard testa stopped water from entering.**

Correct answer: D. Since 7 of the 8 seeds germinated once the coat was scratched, they were alive (viable) but dormant. The hard testa blocked water uptake, and scarification let water in. Non-viable seeds would not germinate at all.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: emergence of maize (Day 7) and beans (Day 8)

Q5. Which statement correctly compares how maize and bean seedlings emerge from the soil?

- A. Maize is hypogeal: its scutellum stays underground while the coleoptile pierces the soil. Beans are epigeal: their two cotyledons are raised above ground.**
- B. Both maize and beans are epigeal because both have a plumule.
- C. Both maize and beans are hypogeal because both are planted 3 cm deep.
- D. Maize is epigeal: its cotyledons are lifted above the ground. Beans are hypogeal.

Correct answer: A. Maize is a monocot with one scutellum that stays below ground (hypogeal), and the coleoptile protects the shoot as it emerges. Beans are dicots whose two cotyledons are lifted above the soil (epigeal).

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: apical dominance (0 vs 3 branches)

Q6. A bean plant with its shoot tip left on grew 0 branches. After the tip was removed at Day 20, it grew 3 branches. What explains this?

- A. Removing the tip stopped the plant from respiring, so the side buds grew.
- B. Auxin made in the shoot tip inhibits the side (lateral) buds, so removing the tip lets them grow into branches.**
- C. Gibberellins from the roots only work after the tip is cut off, so the side buds die.
- D. Absciscic acid from the tip makes the plant grow taller, so branches cannot form.

Correct answer: B. This is apical dominance. Auxin from the apical bud suppresses lateral buds, and when the tip is removed this inhibition stops, so branches grow (cytokinins promote lateral bud growth).

USE AFTER: MODEL BUILDING PHASE — Model revision: comparing Day 1 drawing with final model (shoot apical meristem, primary growth)

Q7. Maize grew to 125 cm and beans to 44 cm by Day 42. In which part of the plant does the primary growth that lengthens shoots and roots take place?

- A. In the apical meristems at the tips of the shoots and roots**
- B. In the testa and micropyle of the seed
- C. In the mature leaf cells that are fully expanded
- D. In the endosperm stored in the seed

Correct answer: A. Primary growth happens where cells divide at the apical meristems at the shoot and root tips. The other parts listed do not make new cells for lengthening.

USE AFTER: MODEL BUILDING PHASE — Community action planning: selling 60 seedlings at KES 20 each (profit calculation)

Q8. The school nursery sells 60 aloe vera-treated seedlings at KES 20 each. The polythene tubes and soil cost KES 300. What is the profit?

- A. KES 900**
- B. KES 1,200
- C. KES 300
- D. KES 1,500

Correct answer: A. Income is $60 \times 20 = \text{KES } 1,200$. Profit is income minus cost: $1,200 - 300 = \text{KES } 900$.

